PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

KIBAHA PRIVATE SECONDARY SCHOOLS ASSOCOATION (KIPSSA)

FORM FOUR PRE – MOCK EXAMINATION – 2022

CHEMISTRY 1

CODE: 032/1

TIME: 3:00 Hours **JUNE 2022**

INSTRUCTIONS

1. This paper consists of section A, B and C
2. Answer all questions in section A and B and only one question from section C
3. Non – Programmable calculate are allowed in Examination room.
4. Write your Examination number on every page of your answer booklet (s)
5. The following constants may be used ;
   Atomic masses H =1, O = 16 S = 32, CL = 35.5, Na = 23, C =12,
   Ag = 108, CU = 63.5, Fe = 56, Zn = 65, Br = 80

   Avegadro's number 6. $02x10^{23}$
   GMV at s.t.p = 22. $4dm^3$
   1 Faraday = 96,500 C
   1 Litre = 1 dm^3 = 1000 cm^3
   Density of water = 1000kg/M^3

This paper consists of Six Printed pages

SECTION A: (15 MARKS)

Answer all questions in this section.

1. For each of the items (i-x), choose the correct answer from among the given alternative and write its letter in the answer booklet provided.

   i. Loose or floppy clothing is not allowed in the laboratory .why?
      A. Movement has to be fast
      B. It will get wet when water splashes.
      C. It may catch fire or cause one to fall
      D. It cause poor ventilation in the body
      E. It prevent experiment from being conducted well

   ii. What mass of sulphuric acid (H_2SO_4) is found in $400cm^3$ of its 0.1M aqueous solution?
      A. 2.67g
      B. 9.8g
      C. 4.89g
      D. 3.92g
      E. 8.69g

   iii. One of the isotopes of an element X has atomic number Z and a mass number A .What is the number of neutrons contained in the nucleus of element?
      A. Z-A
      B. A
      C. A+Z
      D. Z
      E. A-Z

   iv. When methane undergoes substitution reaction with excess chlorine, what is the final product?
      A. Chlorometane
      B. Dichloromethane
      C. Trichloromethane
      D. Tetrachloromethane
      E. Chloroethane

   v. Aluminum does not react with water and does not corrode much in air why?
      A. It is below hydrogen in reactivity series
      B. It forms a stable carbonate with prevent reactions
      C. The metal is covered with protective coating of oxide
      D. It is very stable
      E. Does not react with water

   vi. Substance X liberates chloride gas from acidified potassium chloride. What is the behavior described by X
      A. Catalysis
      B. An oxidation
      C. Reduction
      D. De colourization
      E. An isotope

   vii. When a burning fuel producer blue colour it means those is ;
      A. Adequate supply of oxygen with production of soot.
      B. In adequate the supply of oxygen without production of soot.
      C. Adequate supply of oxygen with production of soot.
      D. Adequate supply of oxygen with production of more heat.

E. In adequate supply of oxygen with production of more heat.

viii. What is the number of molecules in 16.8dm^3 of chlorine at S.T.P?
A. 4.52 X 10^{23} molecules
B. 6.02 X 10^{23} molecules
C. 7.5 X 10^{23} molecules
D. 3.14 X 10^{23} molecules
E. 9.03 x 10^{23} molecules

ix. The chemical techniques which could be used to separate iodine crystal from a mixture of iodine and sand is
A. Evaporation
B. Recrystallization
C. Sublimation
D. Distillation
E. Solvent extraction

x. Which of the following warning signs is likely to be found on the bottle containing concentrated sulphuric acid?
A. Corrosive
B. Explosive
C. Flammable
D. Irritant
E. Oxidizing agent

2. Match the items in list A with the correct response in list B by writing letter of the correct response beside the number in the answer booklet provided.

LIST A	LIST B
i. A black solid element which burns with reddish glow giving off colorless gas which is slightly acidic	A. Calcium B. Nitrogen C. Aluminium
ii. Silvery white metal which burns with golden yellow flame giving an oxide which is basic in nature	D. Iron E. Carbon F. Sodium
iii. A yellow element in colour which burns with blue flame giving colourless gas which is strongly acidic in nature	G. Copper H. Beryllium I. Argon
iv. A shinning white strip metal which burns with dazzling white flame giving an oxide which is slightly basic in nature	J. Magnesium K. Sulphure L. Phosphorus
v. A sliovery white metal which burns with brick red flame giving off oxide which is white solid	M. Chroline N. Zinc O. Fluorine P. Lead

SECTION B (70 MARKS)
Answer all question in this section

3. (a) Consider the following elements.
$^{16}_{8}$O, $^{19}_{9}F$, $^{20}_{10}Ne$, $^{23}_{11}Na$ and $^{24}_{12}Mg$. Atoms and ions of these elements can be iso-electronic (have the same number of elements).
i. Write down their symbols when they are in iso-electronic
ii. Write down their common electron arrangement in their ions and atoms.

(b) Although sulphur dioxide is oxide it can be further oxidized.

i. Write an equation and conditions for further oxidized of sulphur dioxide.
ii. Write down an equation showing how product can be used industrially to obtain a desire material to be produced .
iii. Give the name for this industrial process.
iv. Write down two uses of the product formed by the industrial process named in (iii) above.

4. (a) i. Why petrol tanks in petrol station are constructed underground?
ii. What would happen in petrol to a well stoppered bottle full of water left in a deep freezer over night? Give reason to your answer.
iii. Why iron not usually used in steam pipes and boilers

(b) Explain by giving reason why
i. Laboratory door is opens outward
ii. Laboratory floor is rough and never polished
iii. Fume chamber is important in the chemistry laboratory
iv. Some chemicals /reagents are stored in brown coloured bottles.

5. (a) Write the chemical formula of the following compound
i. Magnesium hydrogen carbonate
ii. Sodium carbonate
iii. Silicon (iv) oxide
iv. Calcium nitrate

(b) Give the reason for your answer. What will happen when
i. Barium chloride solution is added to an aqueous solution of potassium sulphate.
ii. Silver nitrate solution is added to aqueous solution of potassium Chloride
iii. Concentrated sulphuric acid is exposed to the atmosphere.

6. (a) Expalin in terms of chemical bonding why;
i. Graphite conduct electricity while diamond does not conduct.
ii. Graphite is used as lubricant and diamond is used as rock cutting substance.

(b). 8.48g of sodium carbonate crystals where made up to $250cm^3$ of Solution. $25cm^3$ of this solutions neutralized $30cm^3$of 0.20M hydrochloric acid. Calculate the number of moles of water of crystallization in solution in Sodium carbonate crystals.

7. (a) Briefly explain the role of the following substance in the extraction of iron.
i. Coke
ii. Limestone

(b) Solid iodine and gaseous hydrogen were kept in a chosen system and

Heated until the following equilibrium was established.

$$H_{2\,(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$$

If water was introduced in the system hydrogen iodine dissolve in it

i. In which direction would the equilibrium position be shifted
ii. What would happen to the production of hydrogen iodine upon the introduction of water in the reaction system?

8. (a) An organic compound was found formed of 42.6% carbon , 7.1% hydrogen and the rest is for bromine .Calculate
   i. It's molecular mass if its vapour density is 70.5.
   ii. Emprical formular
   iii. Molecular formular
   iv. Write the structure and names of all the possible isomer of the compound formed above.

   (b) In the reactions below,state whether the substances underlined are undergoing oxidation or reduction.

   i. $H_2S + Cl_2 \longrightarrow 2HCL + S$

   ii. $Mno_2 + 4HCL \longrightarrow Mncl_2 + 2H_2O + CL_2$

   iii. $\underline{CU^{2+}} + 2e^- \longrightarrow CU$

9. (a) Difine the following terms
   i. Mole
   ii. Moler volume

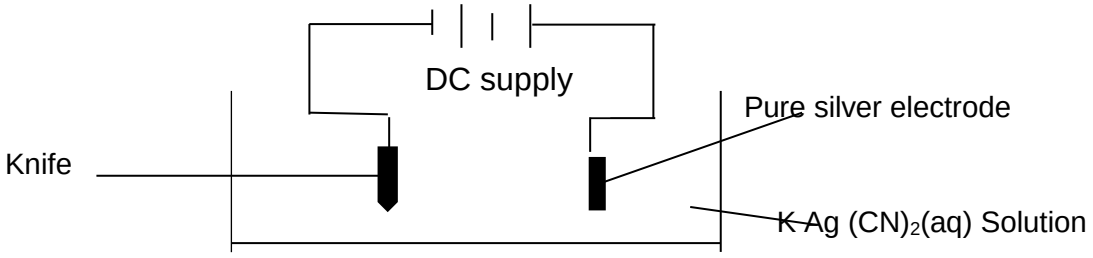
   (b) What volume of carbondioxide measured at S.T.P will be produced when

   21.0g of sodium hydrogen carbonate is completely decomposed? Chemical

   Equation is necessary.

10. (a) A molar solution of ethanoic acid conduct electricity but not easily as a molar Solution of hydrochloric acid. What explanation can you offer for this fact?

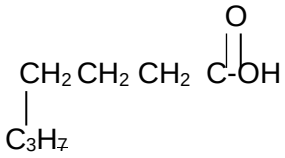
    (b) The apparatus shown below was used in an experiment of electroplating an iron knife with silver .Sample of impure silver of mass of 3.45g was used up of pure gold of mass 6.45g .After the electrolysis the cathode was found to weight 9.66g.
    i. calculate the number of coulumbs of electricity passed
    ii. Find out the percentage purity of the impure silver
    iii. Write equations to represent the reaction happening at the electrodes.
    iv. At which electrode reduction taken place?

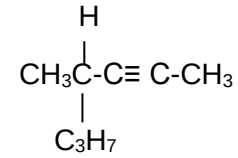


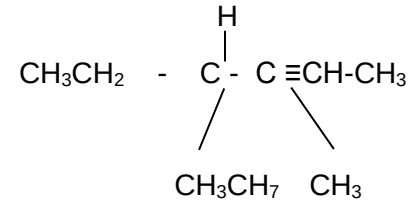
11. (a) Briefly explain any application of neutralization reaction.
    (b) Write examples give two criteria used in classification of fuels.

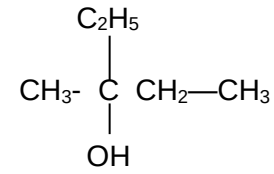
12. (a) i. Name two sources of organic compounds

ii. The following are some organic compounds

A. 

B. 

C. 

D. 

Name each of the compounds and state the family of organic compounds to which each belongs.

(b) i. Write an equation to show the reaction between ethanoic acid and ethanol
ii. What type of reaction taken in b(i) above
iii. Give the name of the main product formed.

SECTION C (15 MARKS)
Answer only one question in this section

13. (a)Explain the effect of urban to empty their poor sewage disposal into sources of water bodies.

(b)List down four causes of ozone layer destruction.

(c) Explain five effects which occur to both animals and plants when ozone Layer is destructed.

14. (a) Explain three factors that influence soil formation

(b) Give four causes of loss in soil fertility.

(c) Give five factors that determine soil productivity and fertility.